

# Majorana zero modes in silicon: quantified failure of the conduction band and a conditional hole-based route

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## Abstract

We assess the feasibility of Majorana zero modes (MZMs) in silicon within a Bogoliubov–de Gennes framework whose validation chain, derivations, parameter tables, and convergence studies are given in the Supplementary Material. Conduction-band wires fail independently on three axes: spin–orbit coupling (best intrinsic topological gap  $\sim 1.5\,\mu\text{eV}$ ; engineered SOC of  $\alpha \gtrsim 0.02\,\text{eV}\,\text{\AA}$  required), a  $g = 2$  ceiling forbidding induced gaps above  $116\,\mu\text{eV}$  at attainable fields (quantifying the optimal-coupling rule of Cole *et al.*), and valley physics—where we prove an exact two-band equivalence for smooth interfaces (Supplement S2) and derive that each single-atomic interface step acts as a near- $\pi$  Josephson junction in the band-projected pairing ( $\Delta\varphi = 2k_0a/4 \approx 0.85\pi$ , Supplement S3), with a numerically determined bound state that caps the topological gap at  $\sim 18\%$  from the very first step. For the valence band we derive the gate-field covariation of  $(\alpha, \hat{g}, m^*)$  from a four-band Luttinger–Kohn fin model: the field that builds direct-Rashba SOC ( $\alpha$  up to  $0.054\,\text{eV}\,\text{\AA}$ , spin–orbit lengths 32–101 nm, matching FinFET measurements) suppresses the wire-axis  $g$  below 1.1 while raising  $g_z$  to 2.4–4.2. Field-orientation maps then select the candidate designs. We label each input by its evidentiary status. *Measured*: Si:B critical fields ( $B_{c2} \approx 0.1\text{--}0.4\,\text{T}$ ), FinFET hole  $l_{so}$  and  $g$ -tensors. *Derived*: the LK operating curve and all topological-gap numbers. *Hypothetical*: Pauli-limited thin-film Si:B and a hard-gap Al/Si-hole interface. On measured inputs only, a thick Si:B parent with tilted field supports  $10\text{--}19\,\mu\text{eV}$ . If  $10\text{--}20\,\text{nm}$  Si:B films reach their Pauli limit (a fabrication target consistent with measured coherence lengths, but not yet demonstrated),  $21\text{--}27\,\mu\text{eV}$ ; if a hard-gap Al/Si-hole interface existed (none does),  $34\,\mu\text{eV}$  at the metallization-renormalized center point and up to  $55\,\mu\text{eV}$  *bare* over field orientations. The hole route is therefore a *conditional* feasibility result, decided by two standard measurements, not an established platform. Transport calculations show our spectral disorder thresholds are conservative, identify an abrupt class-D breakdown mode, and warn that step-disordered wires become Anderson insulators whose hard-looking transport gaps mask degraded local protection. Quasiparticle poisoning restricts the measured-field all-silicon stack to detection (not parity-coherent) experiments unless the parent gap is engineered. Two measurements decide the platform: the parallel critical field of thin Si:B films, and the wire-axis hole  $g$ -tensor in device geometry.

## 1 Introduction

Semiconductor–superconductor hybrid nanowires remain the most studied route to MZMs [1, 2, 3], and a decade of experiments—from the first InSb signatures [4] through the disorder-driven retractions and reappraisals [5] to protocol-based standards [6]—has made one lesson clear: material quality and honest accounting of trivial explanations dominate the problem. Silicon, the most controllable semiconductor in existence, has been conspicuously absent from this program, for reasons

usually stated qualitatively: weak spin–orbit coupling (SOC), a small  $g$ -factor, and conduction-band valley degeneracy.

This paper makes those reasons quantitative, finds that one of them hides a qualitatively new failure mechanism, and shows that none of them applies to silicon *holes*. Our contributions: (i) a quantified electron verdict (Secs. 3.1, 3.2); (ii) an exact unitary equivalence making smooth-interface valley splitting benign under the physical inter-valley pairing channel, and the identification of single-atomic interface steps as near- $\pi$  Josephson junctions that destroy protection from the first step (Sec. 3.3); (iii) a Luttinger–Kohn (LK) derivation of the gate-field covariation of hole parameters, which erodes free-parameter optimism but preserves a viable design (Sec. 4); and (iv) orientation- resolved feasibility maps for two silicon-compatible parents—laser-doped superconducting Si:B [18, 19, 21, 22] and Al films—including thin-film design targets, metallization renormalization, and quasiparticle-poisoning limits (Sec. 4.2). Throughout, spectral diagnostics are cross-checked by recursive-Green’s-function transport (Sec. 5).

## 2 Model and methods

We use the standard single-channel BdG Hamiltonian

$$H = \left( \frac{\hbar^2 k^2}{2m^*} - \mu + V(x) \right) \sigma_0 + \alpha k \sigma_z + E_Z \sigma_x \quad (\text{normal part}), \quad \Delta i \sigma_y \quad (\text{pairing}), \quad (1)$$

discretized with exact particle–hole symmetry by construction, topological for  $E_Z^2 > \Delta^2 + \mu^2$ . Validation (repository Fig. 1): gap closing at the analytic boundary; end-localized zero modes; exponential length splitting; finite-wire vs analytic bulk-gap agreement. Valley physics uses an  $8N$  basis with *inter-valley* singlet pairing  $\Delta(\nu_x \otimes i \sigma_y)$ —the physical zero-momentum channel since the relevant valleys are time-reversed partners—with arbitrary complex valley-orbit (VO) profiles  $\lambda(x)$  and a gauge-covariant phase-locked Dresselhaus-type SOC option. Hole parameters derive from a four-band LK model of the fin cross-section (validated against exact [100] bulk masses) with vertical field, bare ( $\kappa$ ) and orbital Zeeman terms. Transport uses a dimension-generic RGF with Ando wave-matched normal leads and an  $E = 0$  reflection invariant sign  $\det r$  (kwant-equivalent; validated against closed-chain spectra to  $10^{-13}$ ). A test suite pins every exact claim.  $\Delta$  is a *static* induced gap (the  $\omega = 0$  limit of the standard tunneling self-energy); parent-gap field suppression enters via  $\Delta_{ind} \leq \Delta_p(B)$  and metallization via the quasiparticle weight  $Z$  [7, 23, 24] (Supplement S6) — we do not solve the dynamical self-energy problem, and the renormalized numbers bound the direction of the omitted physics. All caricature choices (GL/Zee-man-linear parent gaps, iid onsite disorder with correlation length  $dx$ , hard-wall LK geometry) are stated where used; complete per-figure parameter tables, seed families, and discretization/length/seed/grid convergence studies are in Supplement S4–S5.

## 3 Conduction band: three independent failures

### 3.1 Spin–orbit shortfall

Maximizing the topological gap over  $(\mu, B \leq 2 \text{ T})$  at measured intrinsic Rashba strengths gives  $0.16 \mu\text{eV}$  (typical,  $\alpha = 10^{-4} \text{ eV \AA}$ ) to  $1.5 \mu\text{eV}$  (optimistic,  $10^{-3}$ ): below dilution-fridge protection scales and destroyed by any realistic disorder. Reaching a robust  $20 \mu\text{eV}$  requires  $\alpha \approx 0.015$  (idealized) to  $0.021 \text{ eV \AA}$  (with parent-gap suppression  $\Delta(B)$ )—an optimistic lower bound given unmodeled orbital and metallization corrections. Micromagnet arrays [8, 9] reach this scale in

principle; a 2025 InAs experiment measured synthetic  $\alpha = 0.022 \text{ eV nm} = 0.22 \text{ eV \AA}$  [10]—an order of magnitude *above* our silicon requirement, though demonstrated in a lighter-mass material with a mature hybrid stack, and with micromagnet fringe-field disorder unmodeled here. “Engineered-electron silicon” is therefore marginal rather than absurd—but the valley physics below stacks on top of it.

### 3.2 The $g = 2$ catch-22

Since  $E_Z(2 \text{ T}) = 116 \mu\text{eV}$  for  $g = 2$ , any induced gap  $\Delta \geq 116 \mu\text{eV}$  can never reach the topological phase at attainable fields; in the SOC-limited regime relevant to engineered devices, increasing interface transparency *reduces* the achievable gap, with an optimal parent gap  $\Delta_0 \approx 98 \mu\text{eV}$  once  $\Delta(B)$  suppression is included. This sharpens, for  $g = 2$ , the weak-coupling optimum identified by Cole, Das Sarma and Stanescu [7]. Pauli-limited parents satisfy the constraint automatically (Sec. 4.2).

### 3.3 Valley physics: an exact equivalence and step junctions

**Equivalence theorem.** With inter-valley pairing and *any* uniform VO phase, the wire maps unitarily onto two decoupled bands at  $\mu \mp |\lambda|$  with pairing  $\pm\Delta$  (proof in Supplement S2; the key step is the exact invariance of the inter-valley pairing under the valley rotation,  $U_\phi D U_\phi^T = D$ ; test-suite tolerance  $10^{-6} \mu\text{eV}$  on every eigenvalue, spot checks  $3 \times 10^{-10} \mu\text{eV}$ , for valley-scalar and phase-locked Dresselhaus SOC alike): smooth interfaces are benign, and valley splitting  $\gtrsim 100 \mu\text{eV}$  buys single-valley topology, as the simple wedge picture suggests.

**Step junctions.** Real interfaces are stepped. That steps shift the VO phase by  $2k_0\delta z$  is established quantum-dot physics [12, 11]; the new statement here is its consequence under inter-valley *pairing*: projecting the pairing onto the local valley bands gives each band an order parameter of phase  $-\phi(x)$  (Supplement S3, exact), so each single-atomic step,  $\Delta\varphi = 2k_0(a/4) \approx 0.85\pi$ , is a Josephson junction accidentally close to  $\pi$  in silicon. The bound-state depth is determined numerically; the transparent short-junction formula  $\Delta_t |\cos(\Delta\varphi/2)|$  tracks it to 10–25% (Supplement Table 1). A controlled comparison (same-sign staircase vs sign-randomized steps of equal magnitude) shows net phase *winding* is irrelevant at realistic densities; the damage is per-step junction physics: a single step binds a subgap state at  $4.0 \mu\text{eV}$  (18% of the  $22 \mu\text{eV}$  clean gap), deepening to a Kitaev domain-wall zero mode at  $\Delta\varphi = \pi$  [Fig. 1(b)]. Ensembles at 50 nm step spacing retain only  $\sim 0.7$ – $2.3 \mu\text{eV}$  (median,  $L$ -dependent upper bounds). Bunched double-height steps ( $\Delta\varphi = 1.7\pi \equiv -0.3\pi$ ) are  $\sim 2\times$  milder—step-bunched vicinal surfaces are preferable—while VO-amplitude suppression near steps [12] worsens the damage; wires laid along step edges avoid it entirely ( $|\cos \chi|$  scaling). Smooth phase gradients are a real but subdominant second channel, suppressed at large valley splitting. Transport (Sec. 5) adds a warning: step-dense wires become Anderson insulators whose end-to-end transport gap *grows* with length ( $0.5/5.6/23 \mu\text{eV}$  at  $L = 1.5/3/6 \mu\text{m}$ ) while localized subgap states persist—a hard-looking transport gap does not certify protection. The falsifiable consequence: induced-gap hardness in proximitized Si *electron* devices should degrade with miscut/step density from the very first step, testable by gap spectroscopy on standard miscut wafer series; holes are immune by construction.

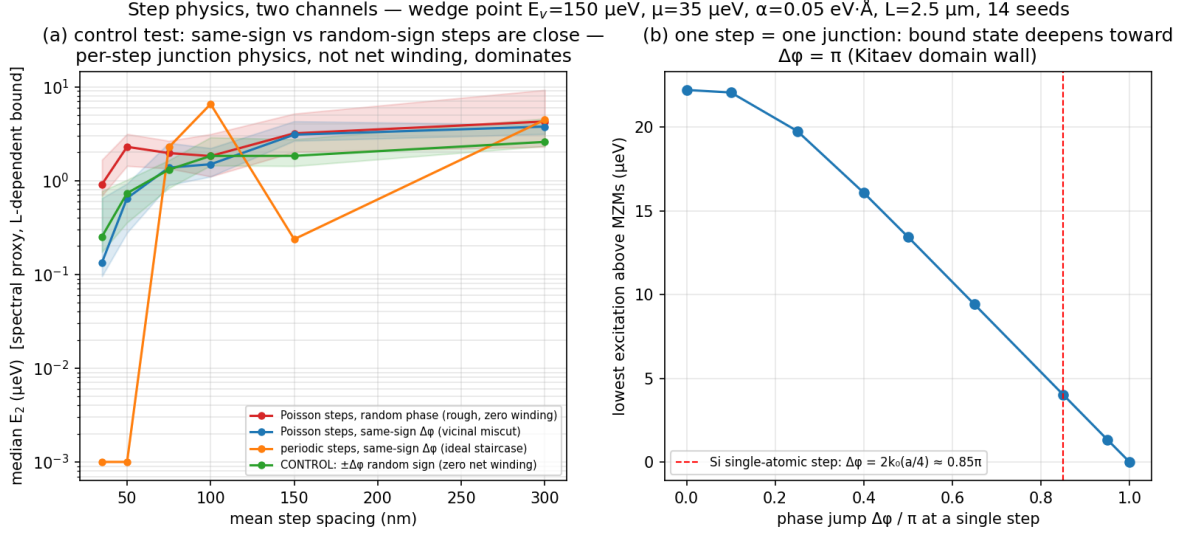


Figure 1: Step physics in the inter-valley-paired electron wire. (a) Controlled scenario comparison at fixed jump magnitude: same-sign (vicinal) and sign-randomized (zero-winding control) staircases are statistically indistinguishable—per-step junction physics, not winding, dominates. (b) A single VO phase step is a Josephson-like junction; the Si single-step value  $0.85\pi$  sits near the Kitaev domain-wall point  $\pi$ .

## 4 Valence band

### 4.1 Luttinger–Kohn constraints: the $\alpha$ – $g$ covariation

Quasi-1D hole channels in Si FinFETs have measured strong, gate-tunable SOC [15, 17, 14] and anisotropic  $g$ -tensors [16]. Our four-band LK fin model reproduces the measured spin–orbit-length window ( $l_{so} = 32\text{--}101 \text{ nm}$ ) and yields the constraint the free parameter box hides [Fig. 2]: the vertical field that builds  $\alpha$  (up to  $0.054 \text{ eV}\cdot\text{\AA}$ ) suppresses the wire-axis  $g_x$  from 1.1 to 0.5 while raising  $g_z$  to 2.4–4.2, with  $m^* \approx 0.44$  and the SOC axis in-plane transverse. A seven-geometry bracket (7–16 nm cross sections, 3–30 MV/m) keeps  $g_x \in [0.4, 1.1]$ : the measured  $g_{xx} \approx 1.9\text{--}2.3$  [16] is *not* reproducible in four-band hard-wall LK, so all platform numbers below are carried under both the LK and empirical tensors; resolving the discrepancy requires six-band  $k \cdot p$  with device electrostatics (top remaining theory item).

### 4.2 Parents, orientation, and the surviving designs

Field-orientation maps [Fig. 3], using the tilted-field topological criterion of Osca, Ruiz and Serra [13], under both  $g$ -tensors and two parents give the design rules: (i) an **Al film** parent is locked to in-plane fields but strong— $34 \mu\text{eV}$  metallization-renormalized at the center point, up to  $55 \mu\text{eV}$  bare over orientations (Fig. 3, not renormalized)—*if* a hard-gap Al/Si-hole interface can be fabricated (none exists); (ii) a **thick Si:B** parent [18, 19, 20, 21] is orientation-tolerant and, using only measured critical fields ( $B_{c2} \approx 0.1\text{--}0.4 \text{ T}$ , orbital-limited), supports  $10\text{--}19 \mu\text{eV}$  at fields tilted toward out-of-plane (riding  $g_z$ ; the center point is  $11.0 \mu\text{eV}$  bare /  $9.8\text{--}10.1$  renormalized on finer optimizer grids, Supplement S5, and moves over  $8.0\text{--}10.6 \mu\text{eV}$  under  $\pm 20\%$  parent-model variations); the in-plane wire-axis direction is  $g$ -starved ( $3\text{--}9 \mu\text{eV}$ ). (iii) The **thin-film Si:B** route turns the “Pauli-limited” hypothesis into a fabrication target: with coherence lengths from measured perpendicular

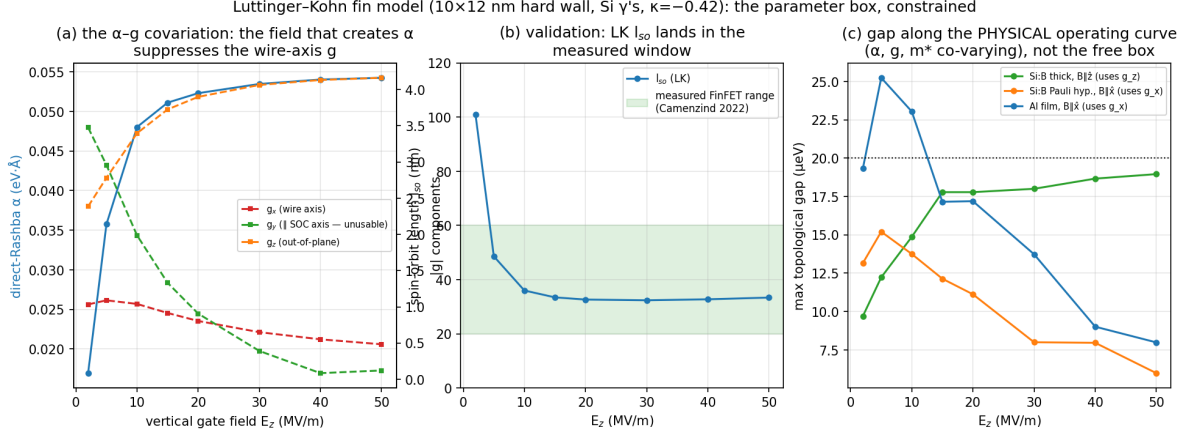


Figure 2: Luttinger–Kohn constraint on the hole platform: (a)  $\alpha$ – $g$  covariation under the gate field; (b) validation of  $l_{so}$  against FinFET measurements; (c) accessible gap along the physical operating curve for three parent/orientation scenarios.

critical fields ( $\xi = 29$ – $57$  nm), films of  $d \lesssim 18$ – $36$  nm have parallel orbital critical fields above the Pauli limit  $B_P = 1.11$  T, and  $d = 10$ – $20$  nm films support **21–27**  $\mu\text{eV}$  (modulo unmodeled  $T_c(d)$  suppression). Quasiparticle poisoning bounds the role of the measured-field stack: at operating fields the Si:B parent gap is 29–33  $\mu\text{eV}$ , so  $x_{qp} = 10^{-6}$  requires  $T_{\text{eff}} \leq 25$ – $29$  mK (Al reference: 130 mK)—a detection platform, not a parity-coherent qubit, absent parent-gap engineering.

## 5 Transport-level verification

RGF transmission with normal leads validates the clean wire (transport gap 37.3 vs bulk 36.8  $\mu\text{eV}$ ) and provides  $\text{sign det } r$  as a topological invariant. Under onsite disorder the spectral proxy (third-smallest  $|E|$ ) is *conservative* at moderate strength (localized states carry no current): the engineered-electron case’s half-gap threshold moves from  $W \approx 410$  to  $\approx 548$   $\mu\text{eV}$ , and breakdown is abrupt—a class-D zero-energy transmission anomaly coinciding with invariant sign flips—rather than a soft closing [Fig. 5]. The same machinery extended to eight-orbital valley cells yields the Anderson-insulator warning of Sec. 3.3.

## 6 Discussion

Two measurements now decide the silicon question. First, the **parallel critical field of thin ( $\leq 20$  nm) laser-doped Si:B films**: if it approaches the Pauli limit, as measured coherence lengths permit, the all-silicon hole stack supports  $\gtrsim 20$   $\mu\text{eV}$  gaps in a CMOS-compatible, same-crystal materials system; if films remain orbital-limited at  $\sim 0.4$  T, the stack is capped near 10–15  $\mu\text{eV}$  and detection-only. Second, the **wire-axis hole  $g$ -factor and SOC axis** in the actual gated geometry: our LK bracket and the published FinFET values differ by  $2\times$  along the one direction that matters for film parents. Both measurements are standard characterization, neither requires a working topological device, and the electron-side miscut prediction offers a third, independent test of the framework. We caution that the step-junction mechanism, the orientation maps, and all platform numbers inherit the stated caricatures (single channel, hard-wall LK, parent-gap models, iid disorder, no orbital magnetic effects [25]); their domains of validity and the sensitivity of every

Field-orientation maps ( $\mu=0$ ): star = in-plane along wire; triangle = out-of-plane.  
 LK says: out-of-plane B ( $g_z$ ) with a thick parent; the in-plane/wire direction is g-starved.

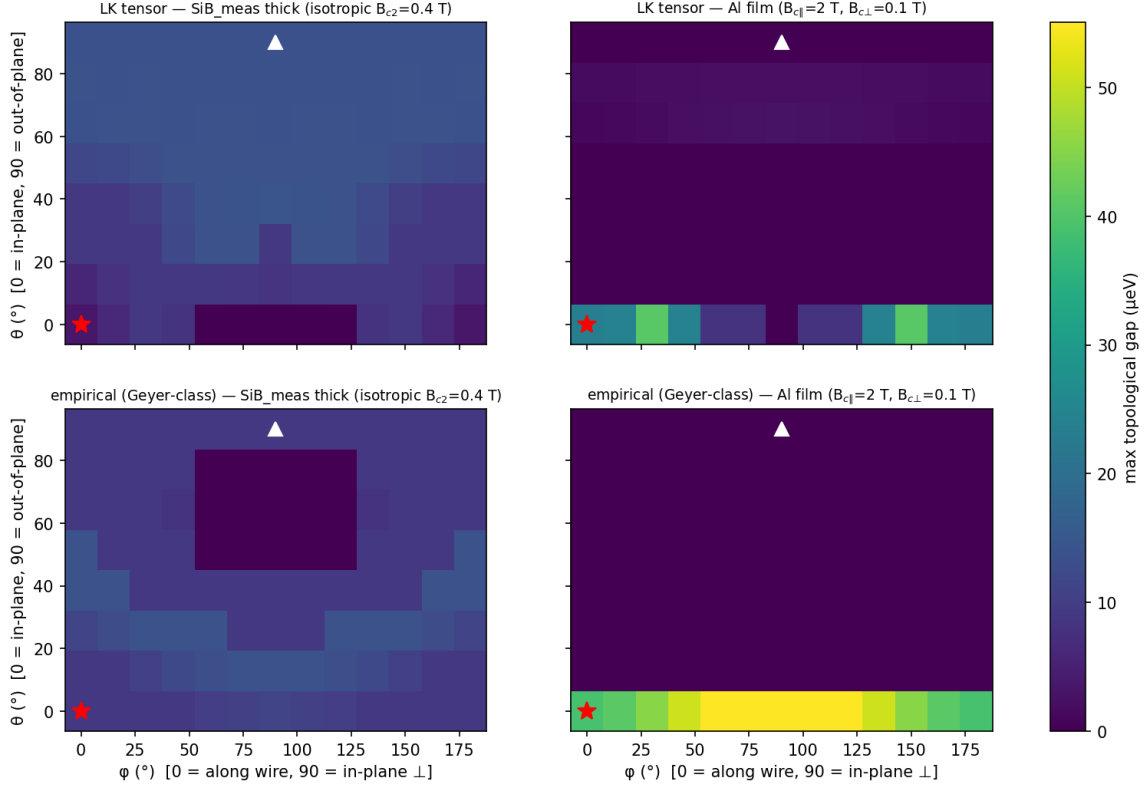


Figure 3: Maximum topological gap vs field orientation for LK (top) and empirical (bottom) hole  $g$ -tensors, with thick measured-Si:B (left) and Al-film (right) parents. Star: in-plane along the wire; triangle: out-of-plane.

headline number to them are quantified in Supplement S5–S6.

## Data and code availability

All figures and quoted numbers regenerate from the committed code (`run_analysis.py`, `transport.py`, `transport_valley.py`, `lk_holes.py`, `qp_poisoning.py`, `convergence.py`, `tests.py`); the numerical ledger is `output/data/key_numbers.json`, and per-figure regeneration commands, the environment specification, and seed families are documented in `README.md` and Supplement S4/S8. An archival snapshot (repository URL, commit hash, and DOI) will be deposited on Zenodo at submission. [Zenodo DOI: to be inserted at submission.]

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Analysis, numerical implementation, verification, and manuscript preparation were assisted by AI agents (Claude, Anthropic) under the author’s direction. Three internal numerical-review rounds and one round of revisions in response to external review are documented, with their corrections, in the repository log. AI-assisted review supplements but does not substitute for the reproducibility

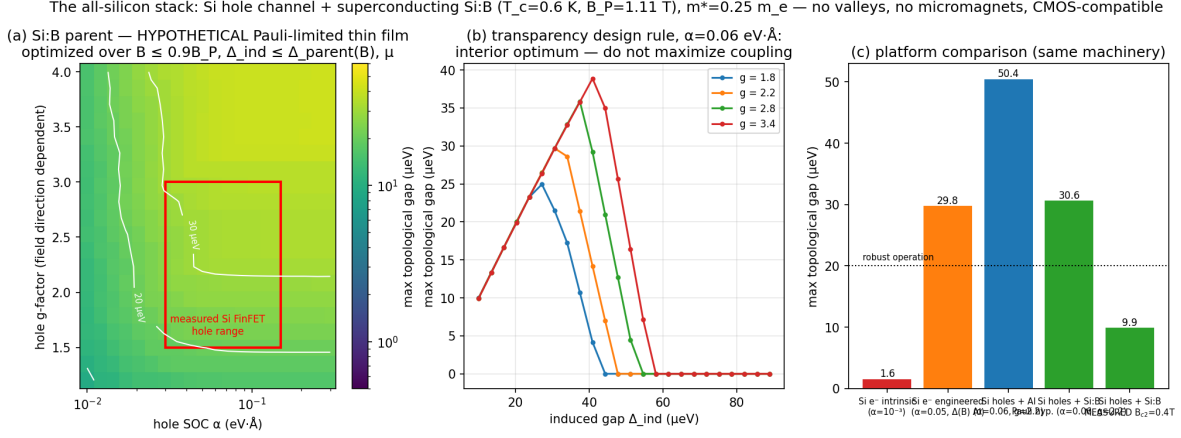


Figure 4: The all-silicon stack under the (hypothetical) Pauli-limited Si:B parent: (a) gap over the empirical  $(\alpha, g)$  box; (b) the transparency design rule (interior optimum); (c) platform comparison including the measured-field Si:B scenario.

materials above; all claims trace to committed, seeded code.

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Fig 12 — transport gap of the engineered-Si wire ( $\alpha=0.05$  eV·Å,  $\Delta=50$   $\mu\text{eV}$ ,  $B=1.5$  T,  $\mu=0$ ,  $L=2$   $\mu\text{m}$ ,  $dx=2.5$  nm; normal leads  $\mu_{\text{lead}}=2000$   $\mu\text{eV}$ )

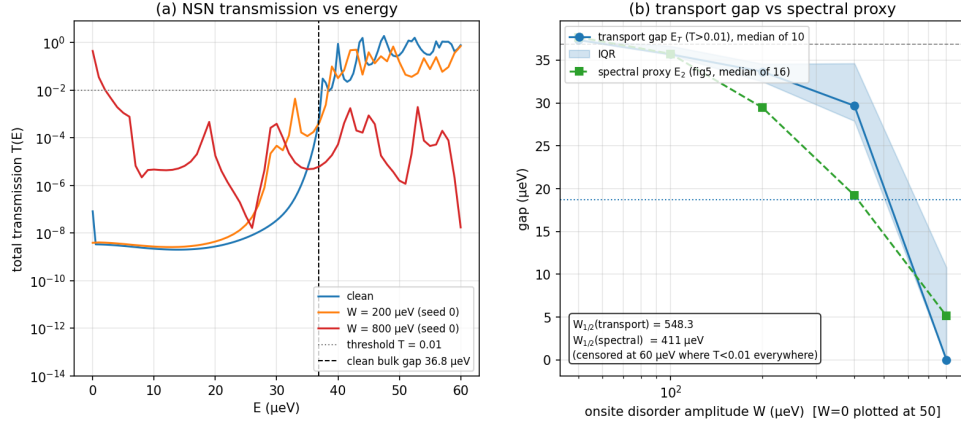


Figure 5: Transport vs spectral disorder thresholds (engineered-electron case): (a) transmission spectra; (b) median transport gap vs disorder with the spectral proxy overlaid.

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